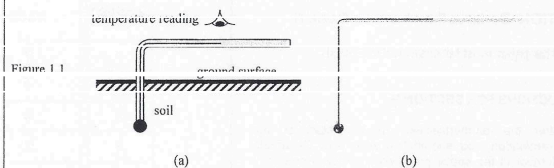


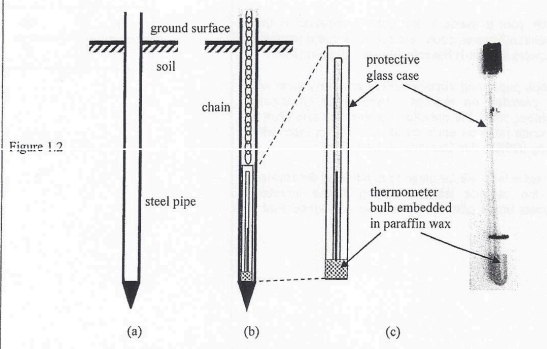
SECTION D: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. Read the following passage about soil thermometer and answer the questions that follow.

The temperature of soil changes with depth, and this information is important to farmers and scientists. To measure soil temperatures at depths close to the ground surface, the bulb of a thermometer is buried in the soil. The stem of the thermometer is bent 90° for easy reading. Figure 1.1a is a schematic diagram and Figure 1.1b shows a photo of a soil thermometer.



For depths greater than 30 cm, a steel pipe is driven into the soil (Figure 1.2a), and a liquid-in-glass thermometer with a protective glass case is lowered into the steel pipe (Figure 1.2b). The bulb of the thermometer is embedded in paraffin wax (Figure 1.2c). To read the temperature, the thermometer is lifted out of the steel pipe by pulling the chain.



Answers written in the margins will not be marked.

Please stick the barcode label here.

(a) As shown in Figure 1.1b, the bulb of the soil thermometer is very large compared to those of common thermometers. Suggest a reason for this design. (1 mark)

(b) On a certain morning, the air temperature is 15°C. An observer takes a measurement of the soil temperature at 1 m deep. The thermometer reading is 20°C. It is given that the mass of the paraffin wax enclosing the thermometer bulb is 0.015 kg, and the specific heat capacity of paraffin wax is $2.9 \times 10^3 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$.

(i) Calculate the energy loss of the paraffin wax as it cools down to the air temperature. (2 marks)

(ii) It is known that the paraffin wax enclosing the bulb of the thermometer gains or loses energy at a constant rate of 0.5 J s^{-1} , estimate the time taken for the paraffin wax to reach the air temperature after the thermometer is lifted out of the soil. (2 marks)

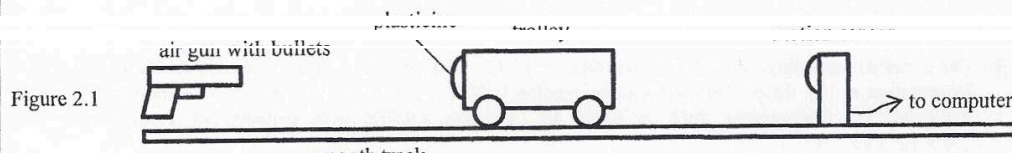
(iii) If there is no paraffin wax enclosing the bulb of the thermometer, explain how the thermometer reading as recorded by the observer is affected. (2 marks)

Answers written in the margins will not be marked.

2. The following experimental items are provided to set up an experiment to estimate the speed of a bullet fired from an air gun.

a smooth track
 a trolley
 a motion sensor used to measure the speed of the trolley
 an air gun and bullets
 an electronic balance

The set-up is shown in Figure 2.1.



Describe the procedures of the experiment. State the physical quantities to be measured and an equation for finding the speed of the bullet. Write down **ONE** precaution for getting a more accurate result. (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Please stick the barcode label here.

*3. The average kinetic energy of one monatomic gas molecule at temperature T is given by

$$E_k = \frac{3}{2} \left(\frac{R}{N_A} \right) T,$$

where R is the universal gas constant and N_A is the Avogadro constant. A monatomic gas is heated from 300 K to 350 K under fixed volume.

(a) Estimate the ratio of the root-mean-square speed ($c_{\text{r.m.s.}}$) of the gas molecules at the two temperatures

$$\left(\frac{c_{\text{r.m.s. at 350 K}}}{c_{\text{r.m.s. at 300 K}}} \right).$$

(2 marks)

(b) Thus, using kinetic theory, explain why the gas pressure would increase.

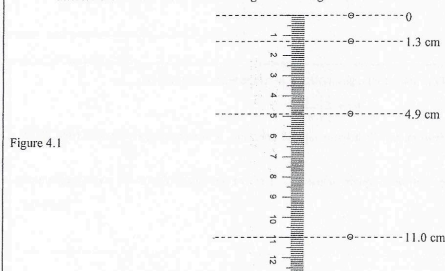
(2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

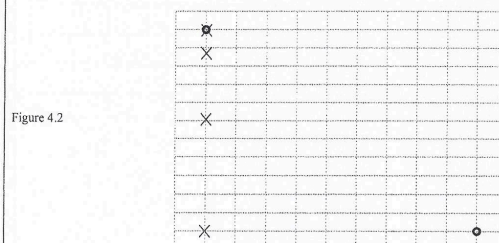
Answers written in the margins will not be marked.

4. (a) A steel ball bearing is released from rest at time $t = 0$. A stroboscopic photo is taken at 0.05 s time intervals. The results are shown in Figure 4.1. Neglect air resistance.



- (i) Estimate the acceleration due to gravity using the data in Figure 4.1. (2 marks)

- *(ii) The bearing is now projected horizontally instead of released from rest. The bearing is projected at time $t = 0$, and a stroboscopic photo is taken at 0.05 s time intervals. The first and the last image of the stroboscopic photo are shown using circles (●) in Figure 4.2. For reference, the stroboscopic photo of the bearing released from rest is also shown in the figure using crosses (x).



Answers written in the margins will not be marked.

Please stick the barcode label here.

- (i) In Figure 4.2, mark the positions of the projected bearing in the stroboscopic photo using circles (●). (2 marks)
- (2) Given that the bearing is projected horizontally with an initial speed of 1 m s^{-1} , use the results of (a)(i) to calculate the speed of the projected bearing when the last image was taken. (3 marks)

- (b) If a small ball is released from rest from the top of a cliff, the speed of the ball becomes constant after a period of time. By considering the forces acting on the ball and using Newton's laws of motion, explain why the speed of the ball becomes constant. (3 marks)

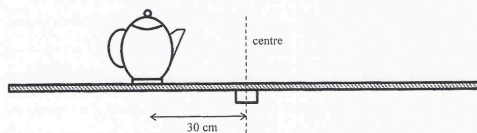
Answers written in the margins will not be marked.

*5. The photo shows a turntable commonly used in restaurants.



A teapot of mass 1 kg is put 30 cm from the centre of a horizontal turntable, Figure 5.1 shows the side view. When the turntable is rotating, the teapot remains at the same position on the turntable.

Figure 5.1



- (a) On Figure 5.1, draw and label all the forces acting on the teapot when the turntable is rotating. (2 marks)
- (b) Taking the teapot as a point mass, estimate the net force acting on the teapot when the turntable is rotating at a rate of 0.5 revolutions per second. (3 marks)

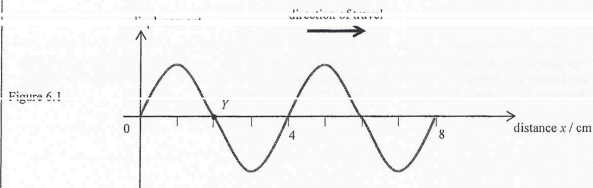
Answers written in the margins will not be marked.

Please stick the barcode label here.

- (c) The turntable is suddenly stopped and the teapot slips. The turntable is rotating at a rate of 0.5 revolutions per second just before it stops, and the frictional force acting on the teapot is 10 N when it is slipping. Determine the distance travelled by the teapot after the turntable stops. (3 marks)

Answers written in the margins will not be marked.

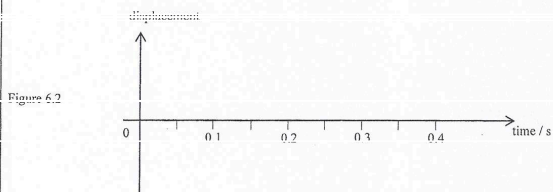
6. (a) A dipper vibrating with a frequency of 5 Hz is put in a water tank. Figure 6.1 shows the displacement-distance graph of the water wave at time $t = 0$. Y is a particle in the water tank.



- (i) Determine the wave speed of the water wave. (2 marks)

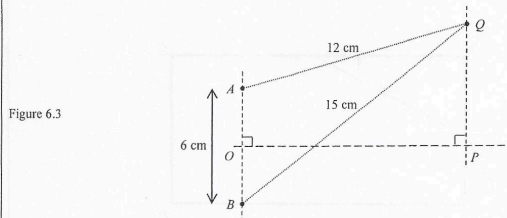
- (ii) State the direction of motion of particle Y at $t = 0$. (1 mark)

- (iii) Sketch the displacement-time graph of particle Y between $t = 0$ and $t = 0.4$ s in Figure 6.2. (2 marks)



Answers written in the margins will not be marked.

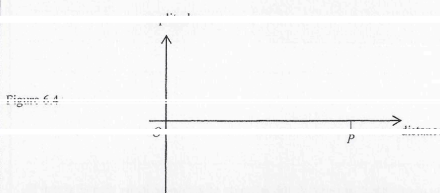
- (b) In Figure 6.3, A and B are two dippers vibrating in phase in a water tank. The distance between A and B is 6 cm. OP is the perpendicular bisector of AB . Q is a second minimum from P , where $AQ = 12$ cm and $BQ = 15$ cm.



- (i) Explain why a minimum occurs at Q . (2 marks)

- (ii) Determine the wavelength of the water wave. (2 marks)

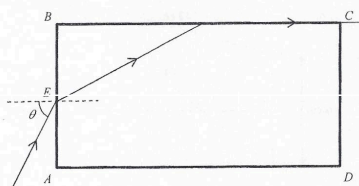
- (iii) Sketch in Figure 6.4 how the **AMPLITUDE** of the water wave varies along the line OP . (1 mark)



Answers written in the margins will not be marked.

7. (a) A light ray enters a rectangular plastic block $ABCD$ from air at point E , and the angle of incidence is θ . The light ray emerges along face BC as shown in Figure 7.1. The refractive index of the plastic is 1.36.

Figure 7.1



- (i) Find the critical angle of the plastic. (2 marks)

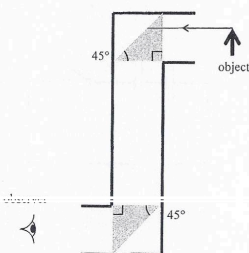
- (ii) Find the value of θ . (3 marks)

- (iii) If the light ray enters the plastic block at point E with an angle of incidence larger than θ , sketch the path of the light ray in Figure 7.1. (2 marks)

Answers written in the margins will not be marked.

- (b) A student designs a periscope using two plastic prisms, the refractive index of the plastic is 1.36. As shown in Figure 7.2, an object is placed in front of the periscope.

Figure 7.2



- (i) Complete the path of the light ray from the object in Figure 7.2, and explain why the periscope fails to work. (3 marks)

- (ii) What can be used to replace the two plastic prisms so that the periscope can work properly? (1 mark)

Answers written in the margins will not be marked.

8. A student uses the following apparatus to measure the resistance of a tungsten filament light bulb.

a battery, a switch, a variable resistor, an ammeter, a voltmeter, a light bulb

- (a) Figure 8.1 shows an incomplete circuit for the experiment. The '+' symbol represents the positive terminal. Use suitable circuit symbols to complete the circuit, and mark the positive terminal of the voltmeter with '+'. (3 marks)

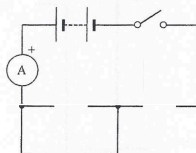
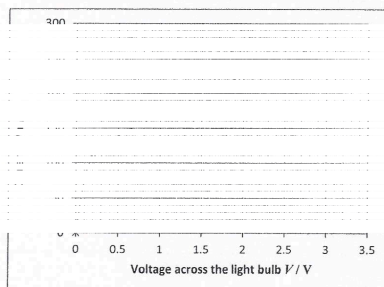


Figure 8.1

The table below and Figure 8.2 show the results obtained.

Voltage across the light bulb V / V	0	0.1	0.2	0.3	0.4	0.5	1.0	2.0	3.0
Current I / mA	0	76	112	126	133	139	170	226	273



Answers written in the margins will not be marked.

- (b) Briefly explain the variation of the resistance of the light bulb with the voltage across the light bulb. (2 marks)

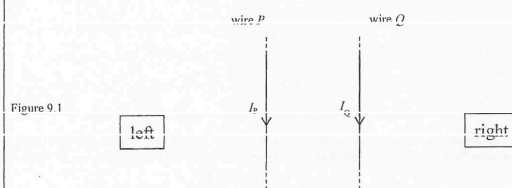
- (c) The student claims that since the resistance of the light bulb is not a constant, the equation $R = V/I$ cannot be used to calculate the resistance of the light bulb. Briefly explain why his claim is wrong. (1 mark)

- (d) Determine the resistance of the light bulb at $V = 0.1 \text{ V}$ and 2.5 V . (3 marks)

- (e) It is given that the cross-sectional area of the tungsten filament in the light bulb is $1.66 \times 10^{-9} \text{ m}^2$, and the resistivity of tungsten at room temperature is about $5.6 \times 10^{-8} \Omega \text{ m}$. Estimate the length of the tungsten filament in the light bulb using the appropriate resistance found in (d). (3 marks)

Answers written in the margins will not be marked.

9. (a) Two long straight current carrying wires, P and Q , are parallel to each other and lie on the plane of the paper as shown in Figure 9.1. The currents in the wires, I_P and I_Q , flow in the same direction.



- (i) State the direction (to the left / to the right / into the paper / out of the paper) of the magnetic field at Q due to P . (1 mark)

- (ii) In Figure 9.1, draw the direction of the magnetic force acting on Q due to P . (1 mark)

- (iii) Show that the magnitude of the magnetic force per unit length F_l acting on Q due to P is

$$F_l = \frac{\mu_0 I_P I_Q}{2\pi r},$$

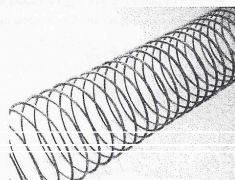
where μ_0 is the permeability of free space and r is the separation between the two wires. (3 marks)

- (iv) For the magnetic force acting on Q due to P and the magnetic force acting on P due to Q , if $I_P \neq I_Q$, briefly explain whether the two forces are equal in magnitude. (2 marks)

Answers written in the margins will not be marked.

- (b) Figure 9.2 shows a metal slinky spring.

Figure 9.2



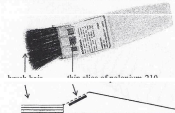
- (i) If a direct current passes through the spring, briefly explain whether the spring will be compressed or stretched due to magnetic force. (2 marks)

- (ii) A student suggests that the spring will be compressed and stretched alternately due to magnetic force when an alternating current passes through. Briefly explain why he is wrong. (1 mark)

Answers written in the margins will not be marked.

10. Dust may adhere to the surfaces of photon and films due to electrostatic attraction. To remove the dust effectively, a special brush with a thin slice of polonium-210 ($^{210}_{84}\text{Po}$) fixed near the brush hair as shown in Figure 10.1 may be used. Polonium-210 undergoes α decay and the daughter nucleus lead ($^{206}_{82}\text{Pb}$) is stable.

Figure 10.1



(a) Write a nuclear equation for the decay of polonium-210. (2 marks)

(b) Briefly explain how the α particles help clean the charged dust. (2 marks)

(c) Briefly explain why the polonium-210 slice must be fixed near to the brush hair. (1 mark)

(d) The manufacturer recommends that the brush should be returned to the factory for replacement of the polonium-210 slice every year. Taking the activity of a newly replaced polonium-210 slice as 1 unit, find its activity after one year (365 days). Given: half-life of polonium-210 is 138 days. (2 marks)

Sources of materials used in this paper will be acknowledged in the booklet *HKDSE Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

Answers written in the margins will not be marked.

List of data, formulae and relationships

Data

molar gas constant	$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
acceleration due to gravity	$g = 9.81 \text{ m s}^{-2}$ (close to the Earth)
universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
speed of light in vacuum	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
charge of electron	$e = 1.60 \times 10^{-19} \text{ C}$
electron rest mass	$m_e = 9.11 \times 10^{-31} \text{ kg}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
atomic mass unit	$u = 1.661 \times 10^{-27} \text{ kg}$ (1 u is equivalent to 931 MeV)
astronomical unit	$AU = 1.50 \times 10^8 \text{ m}$
light year	$ly = 9.46 \times 10^{15} \text{ m}$
parsec	$pc = 3.09 \times 10^{16} \text{ m} = 3.26 ly = 206265 AU$
Stefan constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-2}$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$

Mathematics

For uniformly accelerated motion:

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

Equation of a straight line: $y = mx + c$

Area of triangle:

$$\text{Surface area of cylinder} = 2\pi rh + 2\pi r^2$$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$\text{Surface area of sphere} = 4\pi r^2$$

$$\sin \theta \approx \theta \text{ (in radians)}$$

For small angles, $\sin \theta \approx \tan \theta \approx \theta$ (in radians)

Astronomy and Space Science		Energy and Use of Energy	
$U = -\frac{GMm}{r}$	gravitational potential energy	$E = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$	illuminance
$P = \sigma A T^4$	Stefan's law	$\frac{Q}{t} = \frac{A(T_h - T_c)}{d}$	rate of energy transfer by conduction
$\left \frac{\Delta x}{x} \right < \left \frac{\Delta \lambda}{\lambda} \right $	Rayleigh criterion (resolving power)	$U = \frac{Q}{d}$	thermal transmittance U-value
		$P = \frac{1}{2} \rho A v^3$	maximum power by wind turbine
Atomic World		Medical Physics	
$\frac{1}{2} m_e v_{\text{max}}^2 = hf - \phi$	Einstein's photoelectric equation	$\theta = \frac{1.22 \lambda}{d}$	Rayleigh criterion (resolving power)
$E_n = -\frac{1}{n^2} \left(\frac{m e^4}{8 h^2} \right) = -\frac{13.6}{n^2} \text{ eV}$	energy level equation for hydrogen atom	$\text{power} = \frac{I}{f}$	power of a lens
$\lambda = \frac{h}{mv}$	de Broglie wavelength	$L = 10 \log \frac{I}{I_0}$	intensity level (dB)
$\theta = \frac{1.22 \lambda}{d}$	Rayleigh criterion (resolving power)	$Z = Z_0$	acoustic impedance
		$\alpha = \frac{I_r}{I_0} \frac{(Z_0 - Z)^2}{(Z_0 + Z)^2}$	intensity reflection coefficient
		$I = I_0 e^{-\mu x}$	transmitted intensity through a medium

A1. $E = mc \Delta T$	energy transfer during heating and cooling	D1. $F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$	Coulomb's law
A2. $E = I \Delta t$	energy transfer during change of state	D2. $F = \frac{Q}{4\pi\epsilon_0 r^2}$	electric field strength due to a point charge
A3. $pV = nRT$	equation of state for an ideal gas	D3. $E = \frac{V}{u}$	electric field between parallel plates
A4. $pV = \frac{1}{3} N m \bar{c}^2$	kinetic theory equation	D4. $R = \frac{\rho l}{A}$	resistance and resistivity
A5. $E_k = \frac{3}{2} kT$	molecular kinetic energy	D5. $R = R_1 + R_2$	resistors in series
B1. $F = m \frac{\Delta v}{\Delta t} = \frac{\Delta p}{\Delta t}$	Newton's second law	D6. $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$	resistors in parallel
B2. $\text{moment} = F \times d$	moment of a force	D7. $P = IV = I^2 R$	power in a circuit
B3. $E_p = mgh$	gravitational potential energy	D8. $F = BQv \sin \theta$	force on a moving charge in a magnetic field
B4. $E_k = \frac{1}{2} mv^2$	kinetic energy	D9. $F = BIl \sin \theta$	force on a current carrying conductor in a magnetic field
B5. $F = F_p$	mechanical waves	D10. $B = \frac{\mu_0 I}{2a}$	magnetic field due to a long straight wire
B6. $a = \frac{v^2}{r} = \omega^2 r$	centripetal acceleration	D11. $\mathcal{E} = \frac{\Delta \Phi}{\Delta t}$	induced e.m.f.
B7. $F = \frac{G m_1 m_2}{r^2}$	Newton's law of gravitation	D12. $\frac{V_s}{V_p} = \frac{N_s}{N_p}$	ratio of secondary voltage to primary voltage in a transformer
C1. $\Delta \theta = \frac{2\pi D}{\lambda}$	diffraction width in single slit	E1. $N = N_0 e^{-\lambda t}$	law of radioactive decay
C2. $d \sin \theta = n \lambda$	diffraction grating equation	E2. $t_{1/2} = \frac{\ln 2}{\lambda}$	half-life and decay constant
C3. $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$	equation for a thin lens	E3. $A = \lambda N$	activity and number of nuclei